

AREZOO RAJABI

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Senior AI/ML Engineer / Data Scientist

specializing in machine learning verification and validation, statistical modeling, AI evaluation, multimodal AI, and production ML systems. 10+ years of experience developing and evaluating machine learning systems and 3+ years building and validating LLM/GenAI applications. Experienced in designing evaluation frameworks, defining performance metrics, extracting and curating large-scale datasets, statistical analysis, benchmarking, annotation, error analysis, and identifying edge-case failures. PhD in Computer Science with research and publications in trustworthy AI, adversarial machine learning, privacy, and robust ML.

AREAS OF EXPERTISE

Model Validation & Verification, ML Evaluation & Benchmarking, Statistical Modeling, Evaluation Frameworks, Multimodal AI, Generative AI, LLM/RAG Evaluation, AI Safety, Trustworthy AI, Production ML Systems

SKILLS

ML Evaluation & Validation:	Statistical Analysis, Experimental Design, Hypothesis Testing, Error Analysis, Failure-Mode Analysis, Edge-Case Testing, Benchmarking, Regression Testing, A/B Testing, KPI & Metric Design
Data & Statistical Modeling:	Statistical Modeling, Dataset Creation, Data Extraction, Data Annotation, Feature Engineering, Classification, Regression, Forecasting, Anomaly Detection
ML / AI:	Supervised and Unsupervised Machine Learning, LLMs Post-Training, RAG Agentic AI system design
Programming:	Python, SQL, R, MATLAB
Data & Engineering:	PySpark, GCP, Hadoop, MLflow, Git/GitHub, CI/CD
LLM Training & Evaluation:	LLM quantization (LoRA, qLORA), LLM post-training (RLHF, DPO), LLM Evaluation Harness, LLM Benchmarking, RAGAS, DeepEval, LLM-as-a-Judge for Synthetic Data Generation, Evaluation and Verification, Prompt Engineering and Evaluation, Robustness Evaluation, Trustworthy AI

EXPERIENCE

Senior ML/AI Engineer

Wells Fargo,

Dec 2022 - Present

SF, CA, USA

- **Enterprise LLM Chatbot – 27,000+ users, \$6M ROI**
 - Built automated LLM evaluation and benchmarking pipelines to measure response quality, relevance, factuality, and hallucination rates across production workloads.
 - Designed curated benchmark datasets and evaluation workflows to systematically compare model and prompt performance and identify failure modes.
 - Developed prompt optimization and experimentation frameworks that improved response quality and reduced hallucinations across enterprise LLM workflows.
 - Implemented offline and human-in-the-loop evaluation frameworks to identify model weaknesses and guide iterative improvements.
 - Designed online and offline quality metrics and monitoring frameworks to track model performance and detect quality degradation in production.
- **Multimodal AI for Suspicious Behavior & Policy Violation Detection – \$1.5M+ ROI**
 - Led development of a production multimodal AI system using Vision-Language Models for enterprise safety and policy-violation detection.

- Designed scalable video data collection, annotation, segmentation, and preprocessing pipelines to create evaluation datasets and validate multimodal model performance, robustness, and edge-case behavior.
- Developed benchmarking and error-analysis workflows to evaluate detection accuracy, robustness, and edge-case behavior across diverse video scenarios.
- Reduced multimodal inference and token costs by 80% through intelligent frame selection and automated video segmentation while preserving detection performance.

- **Model Validation and Verification**

- Designed end-to-end evaluation and validation frameworks for production ML and AI systems, defining performance metrics, validation criteria, test datasets, and failure-analysis methodologies for anomaly detection, forecasting, conversational AI, and multimodal models.
- Developed data extraction, curation, annotation, benchmarking, and statistical analysis pipelines to evaluate model performance, reliability, robustness, and edge-case behavior.
- Conducted quantitative model validation through statistical and error analysis, threshold optimization, regression testing, and comparative experimentation to identify failure modes and guide model improvements.

Postdoctoral Scholar

NSL Lab, University of Washington

Feb 2021 - Dec 2022

Seattle, WA, USA

- **Multi-Domain Trojan Detection:** Proposed a multi-domain Trojan detection framework across text, image, and audio models, achieving 85% detection accuracy through extensive benchmarking under adversarial settings.
- **Privacy-Preserving RL Algorithm:** Developed and evaluated privacy-preserving reinforcement learning algorithms, conducting extensive experiments to quantify privacy-utility tradeoffs under adversarial settings.
- **Federated Learning for Trojan Prevention:** Proposed a federated learning approach to counteract Trojan samples during the training phase.

EDUCATION

Ph.D. in Computer Science, Oregon State University 2014 - 2021

Master of Computer Engineering, Sharif University of Technology 2011 - 2013

Bachelor of Computer Science, Sharif University of Technology 2005 - 2010

SELECTED PUBLICATION

1. **A. Rajabi**, S. Asokraj, F. Jiang, L. Niu, B. Ramasubramanian, J. Ritcey, R. Poovendran, MDTD: A Multi-Domain Trojan Detector for Deep Neural Networks, ACM Conference on Computer and Communications Security (ACM CCS), Sep. 2023.
2. J. Jia, Z. Yuan, D. Sahabandu, L. Niu, **A. Rajabi**, B. Ramasubramanian, B. Li, R. Poovendran, FLGAME: A Game-theoretic Defense against Backdoor Attacks In Federated Learning, Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), Sep 2023 (<https://neurips.cc/virtual/2023/poster/70499>).
3. **A. Rajabi**, D. Sahabandu, L. Niu, B. Ramasubramanian, R. Poovendran, LDL: A Defense for Label-Based Membership Inference Attacks, ACM Asia Conference on Computer and Communications Security (AsiaCCS), July 2023 (**7% acceptance rate**).
4. **A. Rajabi**, B. Ramasubramanian, A. Marruf, R. Poovendran, Privacy Preserving Reinforcement Learning Beyond Expectation, Accepted in 61st IEEE Conference on Decision and Control, 2022.
5. **A. Rajabi**, M. Abbasi, R. B. Bobba, K. Tajik, Adversarial Images Against Super-Resolution Convolutional Neural Networks for Free, Privacy Enhancing Technology Symposium (PETS), 2022.
6. M. Abbasi, **A. Rajabi**, C. Shui, C. Gagné, R. B. Bobba, Toward Adversarial Robustness by Diversity in an Ensemble of Specialized Deep Neural Networks, Canadian Conf on Artificial Intelligence, 2020. (Best Paper Award)